

AUDIO NOISE LIMITER

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Audio noise can be annoying, especially if you are trying to listen to a very weak radio station. Peaks of unwanted background noise completely swamp the broadcast signal, making it unintelligible. The circuit presented here overcomes this problem by limiting the noise peaks. It can be used with AM short-wave (SW) receivers and amateur radio (HAM) transceivers.

Circuit and working

Fig. 1 shows the circuit diagram of the audio noise limiter. It is built around TL072 (IC1) and a few other components. TL072 contains two

PARTS LIST

Semiconductors:	
IC1	- TL072 operational amplifier
D1-D2	- 1N4148 signal diode
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R2, R4, R5	- 10-kilo-ohm
R3	- 470-kilo-ohm
R6	- 220-ohm
R7	- 4.7-kilo-ohm
VR1	- 10-kilo-ohm potentiometer
Capacitors:	
C1, C8	- 10 μ F, 35V electrolytic
C2	- 150pF ceramic disk
C3	- 100nF ceramic disk
C4-C6	- 2.2nF ceramic disk
C7	- 220 μ F, 35V electrolytic
Miscellaneous:	
CON1	- 2-pin terminal connector
JACK1, JACK2	- 3.5mm audio jack socket
S1	- On/off switch
	- 12V regulated power supply/12V battery

transistor (JFET) input operational amplifiers (A1 and A2).

Both the noise and the audio signal are fed to the inverting input of A1 (pin 2) in IC1 via capacitor C3. The operational amplifier (or op-amp) is biased by resistors R1 and R2 to half the supply voltage. Its gain is around 100, which helps in buffering the audio signal. Diodes D1 and D2 conduct on noise peaks and limit its peak-to-peak swing to about 1.2 volts.

The next stage is built around op-amp A2. It forms a Sallen-Key Butterworth active low-pass filter. Its cut-off frequency of 5kHz is determined by capacitors C4, C5

and C6. It restricts the audio response and filters out unwanted noise by attenuating frequencies higher than 5kHz. Resistor R6 and capacitor C7 give a clean supply voltage to the op-amps.

Output from potentiometer VR1 can be used to drive LM386 or TDA2003 audio amplifier chips. This greatly reduces the noise.

Changing values of capacitors C4, C5 and C6 to 4.7nF will limit the audio bandwidth to 2.5kHz and further reduce the noise. Due to 4.7k (R7) resistor be-

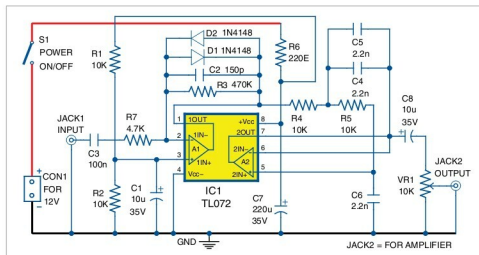


Fig. 1: Circuit diagram of the audio noise limiter

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